ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Symbol reference (used throughout):

- l — antenna physical length (m)
- $\lambda = c/f$ — wavelength (m); $c = 3\times10^8$ m/s
- I_0 — peak current at antenna feed (A)
- R — distance from antenna to observation point (m)
- θ — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377\,\Omega$ — intrinsic impedance of free space
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- $S(R, \theta)$ — average power density radiated (W/m^2)
- $F(\theta) = S(\theta)/S_{\max}$ — normalized radiation intensity (unitless)
- β — half-power beamwidth (rad or deg)
- Ω_p — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity (unitless, often dB)
- $A_e = \lambda^2 D/(4\pi)$ — effective aperture area (m^2)
- A_p — physical cross-section area (m^2)
- G_t, G_r — antenna gains (linear)
- P_t, P_r — transmitted / received power (W)

Key shortcuts: *Hertzian dipole ($l/\lambda < 1/50$) $\Rightarrow D = 1.5$. Half-wave dipole ($l = \lambda/2$) $\Rightarrow D = 1.64$, $R_{rad} = 73\,\Omega$. 3 dB $\Rightarrow$ linear gain of 2.*

Problem 1 — Hertzian Dipole, Power Density at 5 km 8 pts

Setup: A 1-m long dipole excited by a 1 MHz current with amplitude $I_0 = 12$ A. Find the average power density radiated at $R = 5$ km, $\theta = 45°$.

DIPOLE TYPE BY LENGTH

Compute first *$\lambda = c/f$, then ratio l/λ*

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave short-cut, $D = 1.64$, $R_{\text{rad}} = 73\,\Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199\,\Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi}\,\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

Power density *$l/\lambda < 1/50$; far field*

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\sin^2\theta = S_{\max}\sin^2\theta\,(\text{W/m}^2)$$

$k = 2\pi/\lambda$ (rad/m); I_0 peak current (A); l length (m); R obs. dist (m); $\eta_0 = 120\pi \approx 377\,\Omega$.

Max direction *broadside* $\theta_{\max} = \pi/2$

Total radiated power *integrate over 4π*

$$P_{\text{rad}} = \frac{\eta_0\pi}{3}\left(\frac{I_0 l}{\lambda}\right)^2\ (\text{W})$$

Directivity *exact* $D_{\text{Hertz}} = 1.5$ (1.76 dB)

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73
$l=\lambda$	2.41	3.82	199

Half-wave is the default "dipole antenna" unless stated otherwise.

Step 1 — Classify the dipole length. Compute the wavelength to compare with l:

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{10^6} = 300 \text{ m}$$

$$\frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \quad \Rightarrow \textbf{Hertzian dipole}$$

The DIPOLE TYPE BY LENGTH section says short ($l/\lambda < 1/50$) means we can use the Hertzian $S(R,\theta)$ formula directly.

Step 2 — Apply the Hertzian power density formula. From HERTZIAN DIPOLE — $S(R,\theta)$:

$$S(R,\theta) = \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\right)\sin^2\theta$$

Step 3 — Plug in numbers. $\eta_0 = 120\pi$, $k = 2\pi/\lambda = 2\pi/300$ rad/m, $I_0 = 12$ A, $l = 1$ m, $R = 5000$ m, $\theta = \pi/4$:

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5\times10^3)^2}\sin^2(\pi/4)$$

Numerator: $(120\pi)(4\pi^2/9\times10^4)(144) = (120\pi)(144)(4\pi^2)/(9\times10^4)$

$$= 120\cdot144\cdot4\cdot\pi^3/(9\times10^4) = 69120\pi^3/(9\times10^4)$$

Denominator: $32\pi^2\cdot2.5\times10^7$. And $\sin^2(\pi/4) = 1/2$.

Working it out cleanly:

$$\boxed{S(5 \text{ km}, 45^\circ) = 1.51\times10^{-9} \text{ W/m}^2}$$

Sanity: a 12 A current at 1 MHz across a 1-m antenna at 5 km radiates 1.5 nW/m^2. The $1/R^2$ falloff dominates: most of the radiated power scatters into space, only a tiny fraction reaches any one observation point.

Problem 2 — Beam Parameters from Normalized Pattern 15 pts

Setup: $F(\theta) = \exp(-20\theta^2)$ for $0 \le \theta \le \pi$ (θ in radians). Find: (a) half-power beamwidth β, (b) pattern solid angle Ω_p, (c) directivity D.

BEAM PARAMETERS — β, Ω_p, D

Step 1 — Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Step 2 — HPBW β *solve* $F(\theta_{1/2}) = 0.5$*, then* $\beta = 2\theta_{1/2}$ *for symmetric beams*

Gaussian shortcut: $F = e^{-a\theta^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$.

Step 3 — Pattern solid angle Ω_p *units: sr*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\, d\theta\, d\phi$$

No ϕ *dep.:* $\Omega_p = 2\pi\int_0^{\pi} F(\theta)\sin\theta\, d\theta$.

Most integrals have NO closed form. Use Wolfram / numerical.

Wolfram: `2*pi*Integrate[F[t]*Sin[t],{t,0,pi}]`

Step 4 — Directivity D

$$D = \frac{4\pi}{\Omega_p} \quad \text{(unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

Alt. with two orthogonal HPBWs: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$.

Step 5 — Gain (if asked) $\xi =$ *rad. eff.*, $0 < \xi \le 1$

$$G = \xi D \qquad G_{\text{dB}} = 10\log_{10} G$$

Lossless / ideal antenna $\Rightarrow G = D$.

(a) Half-power beamwidth β (5 pts). HPBW is the angular width between the two points where $F(\theta) = 1/2$. Solve $\exp(-20\theta^2) = 0.5$:

$$-20\theta^2 = \ln(0.5) \;\Rightarrow\; \theta = \sqrt{-\ln(0.5)/20} = \sqrt{0.693/20} = \pm 0.186 \text{ rad}$$

The beam is symmetric about $\theta = 0$, so the full beamwidth is:

$$\boxed{\beta = 2\theta = 0.372 \text{ rad} = 21.31^\circ}$$

(b) Pattern solid angle Ω_p (5 pts). From BEAM PARAMETERS:

$$\Omega_p = \iint_{4\pi} F(\theta)\sin\theta\, d\theta\, d\phi = \int_0^{2\pi}\int_0^{\pi} \exp(-20\theta^2)\sin\theta\, d\theta\, d\phi$$

F has no ϕ dependence, so $\int_0^{2\pi} d\phi = 2\pi$:

$$\Omega_p = 2\pi\int_0^{\pi} \exp(-20\theta^2)\sin\theta\, d\theta$$

This integral has no closed form, evaluate numerically (Python/MATLAB/calculator):

$$\boxed{\Omega_p = 0.156 \text{ sr}}$$

(c) Directivity D (5 pts). From BEAM PARAMETERS:

$$D = \frac{4\pi}{\Omega_p} = \frac{4\pi}{0.156}$$

$$\boxed{D = 80.55}$$

Sanity: $D \approx 80$ is highly directive (much narrower than half-wave dipole's 1.64). The narrow Gaussian-like beam ($\beta \approx 21°$) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 — $\lambda/4$ Dipole Radiation Pattern 15 pts

Setup: Dipole of length $l = \lambda/4$. Find: (a) directions of maximum radiation, (b) expression for $S_{\max}$, (c) plot of normalized pattern $F(\theta)$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R *any l*

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

Quarter-wave $l = \lambda/4$ *$\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$*

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4}\cos\theta) - \frac{\sqrt{2}}{2}}{\sin\theta}\right]^2, \quad S_0 = \frac{15I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0\left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858\, S_0$$

Half-wave $l = \lambda/2$ *$\pi l/\lambda = \pi/2$, shortcut form*

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\frac{\cos^2(\frac{\pi}{2}\cos\theta)}{\sin^2\theta}$$

$D = 1.64$, $R_{\text{rad}} = 73\,\Omega$, $\theta_{\max} = \pi/2$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

Half-wave is the default "dipole antenna" unless stated otherwise.

General arbitrary-length dipole pattern. From ARBITRARY-LENGTH DIPOLE — $S(\theta)$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

Plug in $l = \lambda/4$, so $\pi l/\lambda = \pi/4$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right) - \cos(\pi/4)}{\sin\theta}\right]^2 = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}}{\sin\theta}\right]^2$$

Let $S_0 = 15I_0^2/(\pi R^2)$ be the constant prefactor.

(a) Direction of maximum radiation (5 pts). Maximizing $S(\theta)/S_0$ for $0 \le \theta \le \pi$. By calculus or numerical optimization, the max occurs at:

$$\boxed{\theta_{\max} = \pi/2 = 90°}$$

(broadside to the dipole axis, same as half-wave). At $\theta = 0$ or π (along axis), $S = 0$.

(b) $S_{\max}$ (5 pts). Evaluate $S(\pi/2)$:

$$S_{\max} = S(\pi/2) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cdot 0\right) - \frac{\sqrt{2}}{2}}{\sin(\pi/2)}\right]^2 = \frac{15I_0^2}{\pi R^2}\left(1 - \frac{\sqrt{2}}{2}\right)^2$$

Numerically $(1-\sqrt{2}/2)^2 = 0.0858$:

$$\boxed{S_{\max} = 0.0858\,\frac{15I_0^2}{\pi R^2}}$$

(c) Normalized pattern $F(\theta) = S(\theta)/S_{\max}$ (5 pts):

$$F(\theta) = \frac{1}{0.0858}\left[\frac{\cos\!\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}}{\sin\theta}\right]^2$$

Bell-shaped curve peaking at $\theta = \pi/2$ with $F = 1$, dropping to 0 at $\theta = 0$ and π. Symmetric about $\theta = \pi/2$. Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.

Sanity: A $\lambda/4$ dipole has a broadside pattern very close to a Hertzian dipole's $\sin^2\theta$ shape. As $l \to 0$ the bracket approaches $\sin\theta$ and $S \propto \sin^2\theta$ (Hertzian). Half-wave $l = \lambda/2$ has slightly narrower beam than the $\lambda/4$ case.

Problem 4 — Effective Area vs. Physical Cross-section 5 pts

Setup: Half-wave dipole at $f = 100$ MHz, wire diameter $d = 2$ cm. Find effective area A_e and compare to physical cross-section A_p.

EFFECTIVE AREA A_e

Definition *antenna's RX cross section (m^2)*

$$A_e = \frac{\lambda^2 D}{4\pi}\ (\mathrm{m}^2)$$

Physical cross section *wire dipole, diameter d*

$$A_p = l\,d = \frac{\lambda}{2}\,d\ \text{(half-wave)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2}\ \text{(unitless)}$$

Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim\lambda$.

COMMON DIPOLE PARAMETERS

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Half-wave	1.64	2.15	73
$l=\lambda$	2.41	3.82	199

Half-wave is the default "dipole antenna" unless stated otherwise.

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3{\times}10^8}{10^8} = 3\ \mathrm{m}$$

Step 2 — Directivity for half-wave dipole. From COMMON DIPOLE PARAMETERS: $D = 1.64$.

Step 3 — Effective area. From EFFECTIVE AREA A_e:

$$A_e = \frac{\lambda^2 D}{4\pi} = \frac{(3)^2(1.64)}{4\pi} = \frac{14.76}{4\pi}$$

$$\boxed{A_e = 1.17\ \mathrm{m}^2}$$

Step 4 — Physical cross-section. A half-wave dipole has length $l = \lambda/2$. Treat as a rectangle of length l and width d:

$$A_p = l\cdot d = \frac{\lambda}{2}\cdot d = \frac{3}{2}(0.02) = 0.03\ \mathrm{m}^2$$

Step 5 — Ratio.

$$\boxed{\frac{A_e}{A_p} = \frac{1.17}{0.03} = 39.2}$$

Sanity: The effective area is 39× larger than the physical cross-section. This is the key insight of antenna theory: a thin wire antenna captures energy from a much larger "effective" patch of the incoming wave than its physical wire would suggest, because the antenna interacts with the field over a region $\sim\lambda$ around itself.

Problem 5 — Friis Transmission, TV Broadcast 5 pts

Setup: Half-wave dipole TX, 1 kW at 50 MHz. RX home antenna with 3 dB gain, distance 30 km. Find received power.

FRIIS TRANSMISSION FORMULA

Far-field link budget *P_t tx, P_r rx, range R*

$$\boxed{P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.
3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

dB ↔ LINEAR

$$G_{\text{dB}} = 10\log_{10} G_{\text{lin}} \qquad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{5\times10^7} = 6 \text{ m}$$

Step 2 — Antenna gains. TX is half-wave dipole, so $G_t = 1.64$ (from COMMON DIPOLE PARAMETERS).
RX gain: 3 dB → linear via $G_{\text{lin}} = 10^{G_{\text{dB}}/10}$:

$$G_r = 10^{3/10} = 2$$

(3 dB always = ×2, this is the cleanest dB conversion to remember.)

Step 3 — Friis formula. From FRIIS TRANSMISSION FORMULA:

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$$

Step 4 — Plug in. $R = 30$ km $= 3\times10^4$ m, $P_t = 10^3$ W:

$$P_r = (1.64)(2)\left(\frac{6}{4\pi(3\times10^4)}\right)^2 (10^3)$$

The path-loss factor:

$$\frac{\lambda}{4\pi R} = \frac{6}{4\pi\cdot3\times10^4} = \frac{6}{3.77\times10^5} = 1.59\times10^{-5}$$

$$\left(\frac{\lambda}{4\pi R}\right)^2 = 2.53\times10^{-10}$$

$$P_r = (1.64)(2)(2.53\times10^{-10})(10^3) = (3.28)(2.53\times10^{-7})$$

$$\boxed{P_r = 8.3\times10^{-7} \text{ W} = 0.83\ \mu\text{W}}$$

Sanity: 1 kW transmitted, 0.83 μW received at 30 km. That's a 90 dB drop, dominated by the $1/R^2$ free-space path loss. Plenty of signal for a TV receiver (typical sensitivity is -100 dBm $= 10^{-13}$ W), so this link closes comfortably.

Note on the solution PDF: the official key wrote "(1.64)(3)" as a typo, but used $G_r = 2$ in the actual arithmetic (since 3 dB → linear = 2). The final answer 8.3×10^{-7} W matches either way.